

Input Validation

TP1 (Phase One) | CSE 360 | Team 53

Overview

This document lists every input field in Phase One and the rules each value must satisfy. Every value is validated **before** it is used. When validation fails, the user sees a specific error message that explains what to correct, and nothing is saved or acted on. When validation succeeds, the input is processed normally.

- First User Account Setup
- User Login Page
- Admin Home
- Update User Account Details

Input Validation Rules

Screen / Feature	Field	Validation Rules
First User Account Setup	Enter Admin Username	Required. 4-32 characters. Must start with a letter (A-Z, a-z). May contain only letters, digits, and the separators - _ . &. Each separator must be followed by a letter or digit, so no doubled separators and no trailing separator.
First User Account Setup	Enter Admin Password	Required. 8-20 characters. Must contain at least one uppercase letter, one lowercase letter, one digit, and one special character. Only ASCII letters, digits, and these specials are allowed: ~ ` ! @ # \$ % ^ & * () _ - + = { } [] \ : ; " ' < > , . ? /. Spaces and non-ASCII characters are rejected. Must match the "Enter Admin Password Again" field.
First User Account Setup	Enter Admin Password Again	Required. Must satisfy the same password requirements as Enter Admin Password. Must match the "Enter Admin Password" field exactly.
User Login Page: Sign in	Username	Max 32 characters, checked before the database is used. No format check, on purpose, so the page does not reveal whether a username exists. The username must match an existing account.
User Login Page: Sign in	Password	Max 64 characters, checked before the database is used. Must match the account's password, or be a valid one-time password issued by an Admin. Shown masked on screen.
User Login Page: Sign in with a one-time password	Password (one-time password)	Accepted only once, and only before its deadline. On acceptance, the user must set a new password that satisfies every password requirement. The one-time password is cleared once the new password is saved. If the user closes the password window without choosing one, nothing changes and the login does not proceed.
User Login Page: Account setup	Invitation code	Required. 1-10 characters. Blank or whitespace-only input is rejected.

Screen / Feature	Field	Validation Rules
Admin Home: Send An Invitation	Email Address	Required. Max 254 characters. Exactly one @. Mailbox name (before the @) is 1-64 characters, made of letters, digits, and . _ % + -, and may not start or end with a period or contain two periods in a row. Domain (after the @) must contain at least one period. Its parts may contain only letters, digits, and hyphens, and may not be empty or start or end with a hyphen. The last part (top-level domain) must be at least 2 letters. An invitation must not already have been sent to this address.
Admin Home: Send An Invitation	Role (drop-down)	Required. Must be chosen from the list: Admin, Contributor, Viewer, or Curator. The user cannot type a value.
Admin Home: Send An Invitation	Expires on (date picker)	Required. Must be picked from the calendar (typing is disabled). Combined with the time below, the deadline must be in the future and no more than 30 days from now.
Admin Home: Send An Invitation	at (HH:mm, 24-hour)	Required. Format HH:mm on a 24-hour clock, with two digits for the hour (e.g., 09:30, 17:05). Leading and trailing spaces are ignored. Anything over 5 characters after trimming is rejected. Combined with the date above, the deadline must be in the future and no more than 30 days from now.
Update User Account Details	Update Username	Required. 4-32 characters. Must start with a letter (A-Z, a-z). May contain only letters, digits, and the separators - _ . &. Each separator must be followed by a letter or digit, so no doubled separators and no trailing separator. Must not already be taken by another account.
Update User Account Details	Update Password	Required. 8-20 characters. Must contain at least one uppercase letter, one lowercase letter, one digit, and one special character. Only the same ASCII letters, digits, and specials listed for the first-user password are allowed. Spaces and non-ASCII characters are rejected.
Update User Account Details	Update First Name	Optional (an empty value is accepted and clears the field). Max 50 characters. May contain only letters, spaces, hyphens, and apostrophes.
Update User Account Details	Update Middle Name	Same rules as Update First Name.
Update User Account Details	Update Last Name	Same rules as Update First Name.
Update User Account Details	Update Preferred First Name	Same rules as Update First Name.
Update User Account Details	Update Email Address	Required. Same rules as the Admin Home email address (max 254 characters, exactly one @, mailbox name of 1-64 characters, valid domain with a top-level domain of at least 2 letters), except that no check for an earlier invitation applies.